

Objective

The objective of this project is to develop a Python-based **Student Grade Calculator** that accepts marks for multiple subjects and calculates the student's total marks and percentage. Based on predefined percentage conditions, the program assigns an appropriate grade. The project also focuses on practicing **conditional statements, arithmetic operations, user input, and input validation**.

Key Features

- Accepts marks for **multiple subjects**.
- Validates marks within the allowed range of **0–100**.
- Calculates the **total marks** obtained.
- Calculates the **overall percentage**.
- Assigns grades based on predefined conditions:
 - **A+** → 90% and above
 - **A** → 80–89%
 - **B** → 70–79%
 - **C** → 60–69%
 - **D** → 50–59%
 - **E** → 40–49%
 - **F** → Below 40%
- Displays the student's **subject-wise marks**.
- Displays the **total, percentage, grade, and performance**.
- Shows whether the student has **passed or failed**.
- Provides an interactive web interface using Streamlit.
- Can be executed locally using **localhost**.

Technologies Used

- **Python** – Main programming language used to develop the grade calculator.
- **Streamlit** – Used to create the interactive web-based interface.

- **Conditional Statements (if, elif, else)** – Used to assign grades based on percentage.
- **Arithmetic Operations** – Used to calculate total marks and percentage.
- **Input Validation** – Used to ensure that marks are within the valid range.
- **Git & GitHub** – Used for version control and project submission.
- **Localhost** – Used to run and test the Streamlit application locally.

```

app.py
1  import streamlit as st
2
3  st.set_page_config(
4      page_title="Student Grade Calculator",
5      page_icon="📊",
6      layout="centered"
7  )
8
9  st.title("📊 Student Grade Calculator")
10 st.write("Enter the student's marks for each subject to calculate the total, percentage, and grade.")
11
12 st.divider()
13
14 # Student details
15 student_name = st.text_input("Student Name", placeholder="Enter student name")
16
17 st.subheader("📊 Enter Subject Marks")
18
19 # Five subjects
20 subject1 = st.number_input("Subject 1", min_value=0.0, max_value=100.0, value=0.0, step=1.0)
21 subject2 = st.number_input("Subject 2", min_value=0.0, max_value=100.0, value=0.0, step=1.0)
22 subject3 = st.number_input("Subject 3", min_value=0.0, max_value=100.0, value=0.0, step=1.0)
23 subject4 = st.number_input("Subject 4", min_value=0.0, max_value=100.0, value=0.0, step=1.0)
24 subject5 = st.number_input("Subject 5", min_value=0.0, max_value=100.0, value=0.0, step=1.0)
25
26 if st.button("Calculate Grade", type="primary"):
27
28     marks = [
29         subject1,
30         subject2,
31         subject3,
32         subject4,
33         subject5
34     ]
35
36     # Calculate total and percentage
37     total = sum(marks)
38     maximum_marks = 500
39     percentage = (total / maximum_marks) * 100
40

```

```

# Assign grade
if percentage >= 90:
    grade = "A+"
    description = "Excellent"
elif percentage >= 80:
    grade = "A"
    description = "Very Good"
elif percentage >= 70:
    grade = "B"
    description = "Good"
elif percentage >= 60:
    grade = "C"
    description = "Average"
elif percentage >= 50:
    grade = "D"
    description = "Pass"
elif percentage >= 40:
    grade = "E"
    description = "Pass"
else:
    grade = "F"
    description = "Fail"

st.divider()

st.subheader("📊 Student Result")

if student_name:
    st.write(f"**Student Name:** {student_name}")

st.write(f"**Total Marks:** {total:.0f} / {maximum_marks}")
st.write(f"**Percentage:** {percentage:.2f}%")
st.write(f"**Grade:** {grade}")
st.write(f"**Performance:** {description}")

# Result message
if grade == "F":
    st.error("❌ Student has failed.")
else:
    st.success(f"✅ Student has passed with Grade {grade}.")

```

```
st.subheader("📊 Subject-wise Marks")
```

```

result_data = {
    "Subject": [
        "Subject 1",
        "Subject 2",
        "Subject 3",
        "Subject 4",
        "Subject 5"
    ],
    "Marks": marks
}

```

```
st.table(result_data)
```

```
st.divider()
```

```
st.subheader("📊 Grade Criteria")
```

```

st.write("""
- **A+** → 90% and above
- **A** → 80%
- **B** → 70%
- **C** → 60%
- **D** → 50%
- **E** → 40%
- **F** → Below 40%
""")

```

```

PS C:\Users\shamb\OneDrive\Desktop\VEDA Internship\DAY 4> python -m streamlit --version
Streamlit, version 1.64.0
PS C:\Users\shamb\OneDrive\Desktop\VEDA Internship\DAY 4> python -m streamlit run app.py
2026-09-23 17:43:41.974 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.4:8501

Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.

Exception in callback _ProactorBasePipeTransport._call_connection_lost()
handle: <Handle _ProactorBasePipeTransport._call_connection_lost()>
Traceback (most recent call last):
  File "C:\Users\shamb\AppData\Local\Python\pythoncore-3.14-64\Lib\asyncio\events.py", line 94, in _run
    self._context.run(self._callback, *self._args)
  File "C:\Users\shamb\AppData\Local\Python\pythoncore-3.14-64\Lib\asyncio\proactor_events.py", line 165, in _call_connection_lost
    self._sock.shutdown(socket.SHUT_RDWR)
    ~~~~~~^~~~~~
ConnectionResetError: [WinError 10054] An existing connection was forcibly closed by the remote host

```

OUTPUT : <http://192.168.1.4:8501>



Student Grade Calculator

Enter the student's marks for each subject to calculate the total, percentage, and grade.

Student Name

Akshara



Enter Subject Marks

Subject 1

85.00

- +

Subject 2

78.00

- +

Subject 3

45.00

- +

Subject 4

67.00

- +

Subject 5

90.00

- +

Calculate Grade



Student Result

Student Name: Akshara

Total Marks: 365 / 500

Percentage: 73.00%

Grade: B

Performance: Good



Student has passed with Grade B.



Subject-wise Marks

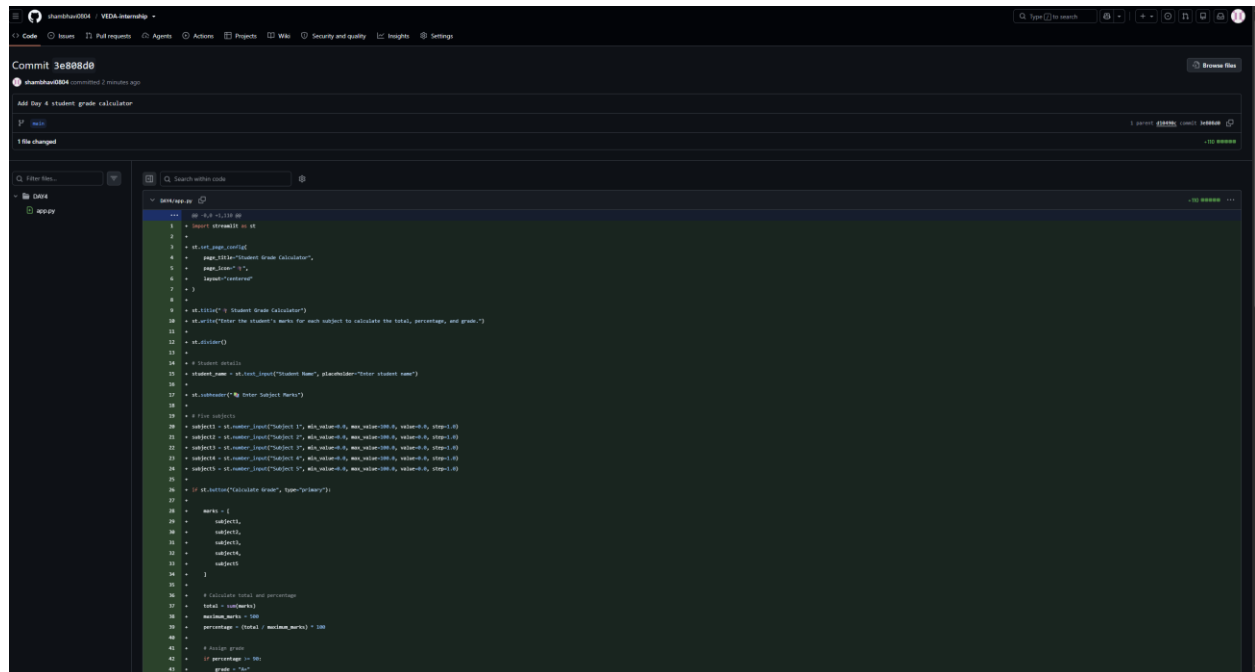
Subject	Marks
Subject 1	85.0000
Subject 2	78.0000
Subject 3	45.0000
Subject 4	67.0000
Subject 5	90.0000



Grade Criteria

- A+ → 90% and above
- A → 80% – 89%
- B → 70% – 79%
- C → 60% – 69%
- D → 50% – 59%
- E → 40% – 49%
- F → Below 40%

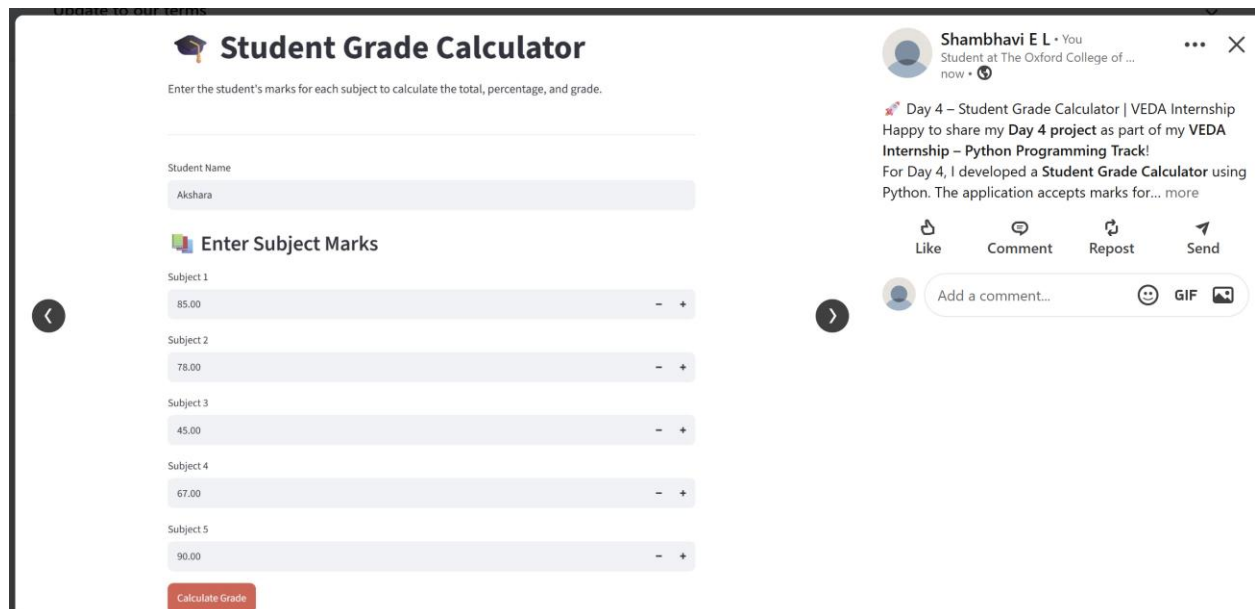
Github: <https://github.com/shambhavi0804/VEDA-internship/commit/3e808d0cfc5ca670ea440104b2b7d7640e4a3c7f>



The screenshot shows a GitHub commit page for the repository 'shambhavi0804 / VEDA-internship'. The commit is titled 'Add Day 4 student grade calculator' and is identified by the hash '3e808d0'. The code is written in Python and is part of a file named 'app.py'. The code defines a Flask application that calculates a student's total marks, percentage, and grade based on five subjects. The user interface is simple, with a form for entering the student's name and marks for five subjects, and a 'Calculate Grade' button.

```
1 from flask import Flask
2
3 # Create app object
4 app = Flask(__name__)
5
6 # Home route
7 @app.route('/')
8 def home():
9     return render_template("index.html")
10
11 # Student details
12 @app.route('/student_details')
13 def student_details():
14     return render_template("student_details.html")
15
16 # Student name
17 @app.route('/student_name', methods=['GET', 'POST'])
18 def student_name():
19     if request.method == 'POST':
20         student_name = request.form['student_name']
21         return render_template("student_details.html", student_name=student_name)
22     else:
23         return render_template("student_details.html")
24
25 # Five subjects
26 @app.route('/five_subjects', methods=['GET', 'POST'])
27 def five_subjects():
28     if request.method == 'POST':
29         subject1 = request.form['subject1']
30         subject2 = request.form['subject2']
31         subject3 = request.form['subject3']
32         subject4 = request.form['subject4']
33         subject5 = request.form['subject5']
34         return render_template("student_details.html", subject1=subject1, subject2=subject2, subject3=subject3, subject4=subject4, subject5=subject5)
35     else:
36         return render_template("student_details.html")
37
38 # Calculate grade
39 @app.route('/calculate_grade', methods=['GET', 'POST'])
40 def calculate_grade():
41     if request.method == 'POST':
42         subject1 = request.form['subject1']
43         subject2 = request.form['subject2']
44         subject3 = request.form['subject3']
45         subject4 = request.form['subject4']
46         subject5 = request.form['subject5']
47         total = int(subject1) + int(subject2) + int(subject3) + int(subject4) + int(subject5)
48         percentage = (total / 500) * 100
49         grade = calculate_grade(percentage)
50         return render_template("student_details.html", subject1=subject1, subject2=subject2, subject3=subject3, subject4=subject4, subject5=subject5, total=total, percentage=percentage, grade=grade)
51     else:
52         return render_template("student_details.html")
53
54 # Calculate grade
55 def calculate_grade(percentage):
56     if percentage >= 90:
57         grade = 'A'
58     elif percentage >= 80:
59         grade = 'B'
60     elif percentage >= 70:
61         grade = 'C'
62     elif percentage >= 60:
63         grade = 'D'
64     else:
65         grade = 'F'
66     return grade
```

https://lnkd.in/p/dn_dCeZb



The screenshot shows a LinkedIn post by Shambhavi E L, a student at The Oxford College of Engineering. The post is titled 'Day 4 – Student Grade Calculator | VEDA Internship' and describes the development of a Student Grade Calculator using Python. The application accepts marks for five subjects and calculates the total, percentage, and grade. The post includes a screenshot of the application's user interface, which features a form for entering the student's name and marks for five subjects, and a 'Calculate Grade' button. The application is running on a web browser, and the user interface is clean and modern.

Student Grade Calculator

Enter the student's marks for each subject to calculate the total, percentage, and grade.

Student Name: Akshara

Enter Subject Marks

Subject 1: 85.00

Subject 2: 78.00

Subject 3: 45.00

Subject 4: 67.00

Subject 5: 90.00

Calculate Grade